

3	(a) (i) 1. amplitude = 1.7 cm	A1	[1]
	2. period = 0.36 cm	C1	
	frequency = $1/0.36$		
	= 2.8 Hz	A1	[2]
	(ii) $a = (-)\omega^2 x$ and $\omega = 2\pi/T$	C1	
	acceleration = $(2\pi/0.36)^2 \times 1.7 \times 10^{-2}$	M1	
	= 5.2 m s^{-2}	A0	[2]
	(b) graph: straight line, through origin, with negative gradient	M1	
	from $(-1.7 \times 10^{-2}, 5.2)$ to $(1.7 \times 10^{-2}, -5.2)$	A1	[2]
	(if scale not reasonable, do not allow second mark)		
	(c) either kinetic energy = $\frac{1}{2}m\omega^2(x_0^2 - x^2)$		
	or potential energy = $\frac{1}{2}m\omega^2 x^2$ and <u>potential energy = kinetic energy</u>	B1	
	$\frac{1}{2}m\omega^2(x_0^2 - x^2) = \frac{1}{2} \times \frac{1}{2}m\omega^2 x_0^2$ or $\frac{1}{2}m\omega^2 x^2 = \frac{1}{2} \times \frac{1}{2}m\omega^2 x_0^2$	C1	
	$x_0^2 = 2x^2$		
	$x = x_0 / \sqrt{2} = 1.7 / \sqrt{2}$		
	= 1.2 cm	A1	[3]